Midterm 2

Put your name here

DATA 412/612

2026-11-11

November 11, 2026. You have 75 minutes. Question 1 is 100 points. Question 2 is 10 points of extra credit.

Put your name in the subtitle at the top of this file. Follow the steps of question 1 in order. Write the code for each step in the chunk under that step. Render to PDF. Submit this `.qmd` and the PDF before the 75 minutes are up.

- You may use your notes and the R help files inside RStudio.
- You may use this course website, https://amousavi1.github.io/612/data-612.html, and David Gerard's data page for the database file, https://dcgerard.github.io/stat_412_612/data.html.
- You may not use Google, any other website, or an LLM.
- You may not communicate with anyone else.
- Use the tidyverse where a tidyverse verb exists. For text, use stringr. Question 2 is two patterns.
- `collect()` before `separate()`, `unite()`, or any `str_` verb. Those run in R. This SQLite file does not run a regular expression.
- Either pipe is fine: `|>` or `%>%`.
- Partial credit is given for code that runs. If you are stuck, go to the next step.

Question 1 (100 points)

The goal is a plot of mean potential against the standard deviation of sprint speed. The players in that plot have a simple name, a first name that starts with M, and a family name of at least 8 letters. A simple name has exactly two uppercase letters and exactly one space.

Each step starts from the data frame the previous step made. Step 8 prints a summary and must not drop any rows. Steps 9 and 10 continue from the data frame in step 7.

1. Load tidyverse, DBI, RSQLite, and dbplyr. This chunk must not display messages or warnings. (5 points)

2. Download `soccer.sqlite`, the European soccer database, from David Gerard's data page, https://dcgerard.github.io/stat_412_612/data.html. Unzip it, and save `soccer.sqlite` in a folder `data` next to this file. Connect with `dbConnect()` and `SQLite()` using the relative path `data/soccer.sqlite`, and assign the connection to `con`. Print the table names, and print how many tables remain when `sqlite_sequence` is not counted. (8 points)

3. Collect `Player` into a data frame. Collect `Player_Attributes` into a data frame, and keep only `player_api_id`, `potential`, and `sprint_speed`. From step 4 on, work with these data frames. (8 points)

4. Prove that `player_api_id` is a primary key of `Player`, and that it is not a primary key of `Player_Attributes`. For each data frame, print the number of rows and the number of distinct `player_api_id`. In one sentence, say what those two comparisons show. (10 points)

5. From `Player`, keep `player_api_id` and `player_name`. (5 points)

6. Keep the simple names. A simple name has exactly two uppercase letters and exactly one space. Use `str_count()` for both counts. (8 points)

7. Split `player_name` on the space into `first_name` and `family_name`, with `separate()`. (6 points)

8. From the data frame in step 7, build `initials` with `str_sub()` and `str_c()`: the first letter of `first_name`, a period, the first letter of `family_name`, and a period. `Luis Martinez` becomes `L.M.`. Show the five initials that occur most often, with their counts, largest count first. Break a tie by the initial, A to Z. Leave the data frame from step 7 intact. (8 points)

9. From the data frame in step 7, keep rows whose `first_name` starts with `M` and whose `family_name` has at least 8 characters. Use `str_starts()` and `str_length()`. (8 points)

10. Unite `first_name` and `family_name` into `player_name`, with one space between them. Keep `player_api_id` and `player_name` only. Assign the result to `df1`, and print how many rows it has. (6 points)

11. From `Player_Attributes`, group by `player_api_id`. For each player, compute `potential_avg`, the mean of `potential`, and `sd_speed`, the standard deviation of `sprint_speed`. Ignore missing values. Assign the result to `df2`. (10 points)

12. Left join `df1` to `df2` by `player_api_id`. Drop a row when `potential_avg` or `sd_speed` is missing. Print how many rows remain. (8 points)

13. Plot `sd_speed` on the horizontal axis and `potential_avg` on the vertical axis. Use points with an alpha so overlapping points stay visible, and one straight smooth in black, dashed, with no gray band. Use `theme_bw()`. Title the plot `Potential and sprint speed`. Label the axes `SD of sprint speed` and `Mean potential`. In one sentence, say how mean potential changes as the standard deviation of sprint speed increases. (10 points)

Question 2 (10 extra credit)

The chunk below is complete. Do not edit it. Write one pattern in each answer chunk.

```
badges <- c(
  "Ab3defgh",
  "k9MnoPqr",
  "A1b2C3de",
  "Abcdefg1",
  "abcdefgh",
  "ABCDEF3H",
  "Ab3 defg",
  "Ab3de",
  "Ab3defghijklmno",
  "A12bcdef",
  "Zz9yyyyy"
)

pairs <- c("abec", "atom", "idea", "kite", "upon")
```

1. Write one pattern and pass `badges` to `str_detect()`. A badge matches only when every condition below is true. (5 points)

- It is 8 to 14 characters long.

- It contains no two digits in a row.
- It does not end with a digit.

2. Write one pattern with two groups and pass `pairs` to `str_replace_all()`. Each group is one vowel followed by one consonant. The two pairs do not have to be the same. Switch the pairs. Vowels are `a`, `e`, `i`, `o`, and `u`. (5 points)